



B5 Organisms exchange with their environment (HL)

B · Form and function · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Multicellular organisms rely on specialised transport and exchange systems because diffusion alone is too slow over large distances. In plants, xylem transports water and minerals from roots to shoots via the cohesion-tension mechanism, while phloem transports sugars from source to sink by mass flow, driven by a pressure gradient created through active loading and unloading of sucrose (pressure flow hypothesis). In mammals, a double circulatory system, with a four-chambered heart separating pulmonary and systemic circuits, allows blood to be re-pressurised after the low-pressure lungs, maintaining high blood pressure for efficient delivery of oxygen and nutrients to body tissues.

Worked example: follow the reasoning

Explaining the pressure flow hypothesis of phloem transport with representative pressures. At the source (e.g. a photosynthesising leaf), companion cells actively load sucrose into the sieve tubes using ATP-powered proton pumps and cotransport, lowering the water potential inside the sieve tube; water then enters by osmosis from the xylem, raising hydrostatic pressure at the source to roughly 2000–2400 kPa. At the sink (e.g. a growing root), sucrose is unloaded and used or stored, raising the sieve tube's water potential there so water leaves by osmosis, keeping sink pressure much lower, e.g. around 500 kPa. Conclusion: this pressure difference (high at source, low at sink) drives a mass flow of phloem sap from source to sink through the sieve tubes.

Quick check · multiple choice

What provides the energy for actively loading sucrose into the phloem sieve tubes at the source?

- A Passive diffusion down a concentration gradient
- B ATP-powered proton pumps enabling active transport/cotransport of sucrose
- C Osmosis alone
- D The transpiration pull generated in the xylem

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A plant physiologist measures hydrostatic pressure of 2200 kPa in the phloem near a mature photosynthesising leaf and 600 kPa in the phloem near a growing root tip on the same plant. [8]
 - (a) Identify which of these two locations is acting as the source and which is the sink, with a reason. [2]
 - (b) Explain, using the pressure flow hypothesis, how this pressure difference causes sucrose-containing sap to move between the two locations. [4]
 - (c) Explain why active loading of sucrose at the source is essential for generating this pressure difference. [2]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) The leaf is the source (higher pressure, 2200 kPa) because it is producing sugar by photosynthesis; the root tip is the sink (lower pressure, 600 kPa) because it is using sugar for growth/respiration.

(b) Active loading of sucrose into sieve tubes at the source lowers the water potential there, so water enters by osmosis from the xylem, raising hydrostatic pressure; at the sink, sucrose is removed (used/stored), raising the sieve tube's water potential so water leaves, keeping pressure low; this creates a pressure gradient from source (2200 kPa) to sink (600 kPa), and the sap (water plus dissolved sucrose) flows en masse down this pressure gradient through the sieve tubes from source to sink.

(c) Without active loading, sucrose could not be concentrated inside the sieve tubes against its own concentration gradient; this active step is what lowers the sieve tube water potential at the source, drawing in water and generating the high hydrostatic pressure that powers the entire mass flow — without it, no pressure gradient (and therefore no net transport) would be established.

Quick check answer

B. Companion cells use ATP to actively pump H^+ ions, creating a gradient that drives cotransport of sucrose into the sieve tube, against its concentration gradient.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.